SHOPSPHERE ENGINEERING

RELEASE NOTES

payment-service 2.4.0 Release Notes

Maintenance release: dependency updates, clearer validation error messages, new request metric

Service	payment-service
Release	2.4.0
Previous version	2.3.1
Release date	2026-10-21
Owner	Payments Team
Release type	Maintenance
Database migrations	None

Fictional document. ShopSphere, its services, people and data are synthetic. This document is part of the TraceIQ training dataset and is not documentation for any real system.

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1. Summary

Version 2.4.0 is a maintenance release. It updates dependencies, improves error messages, and adds request metrics.

1.1 Release Information

Item	Value
Service	payment-service
Release	2.4.0
Release date	2026-10-21
Previous version	2.3.1
Owner	Payments Team
Release type	Maintenance release
Request schema	payment-request/2.0 (Payment API v2)
Database migrations	No database migrations
Breaking changes	No breaking changes to the public payment API are expected
Known issues	None at release time

Table 1-1. Release information.

1.2 Version Path

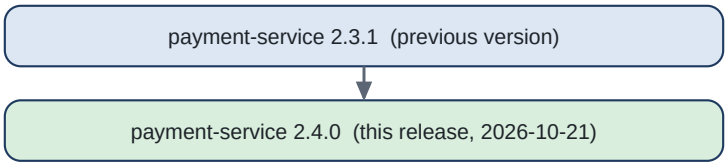

Figure 1-1. Upgrade path covered by these notes.

1.3 Changes at a Glance

Area	Change
Dependency updates	Upgraded request-validation library schemaguard from 2.8.4 to 3.0.1.
Dependency updates	Upgraded HTTP framework to the latest patch version.
Request handling	Request schema handling aligned with Payment API v2 (schema payment-request/2.0).
Request handling	Validation error messages now list every failing field instead of only the first one.
Observability	New metric payment_validation_failures_total, labelled by error code.

Table 1-2. All changes in 2.4.0.

2. Changes

This section lists every change in 2.4.0, grouped as in the original release notes.

2.1 Dependency Updates

- Upgraded request-validation library **schemaguard** from 2.8.4 to 3.0.1.
- Upgraded HTTP framework to the latest patch version.

Dependency	Role in payment-service	2.3.1	2.4.0
`schemaguard`	Request-validation library	2.8.4	3.0.1
`HTTP framework`	HTTP framework	previous patch version	latest patch version

Table 2-1. Dependency versions before and after.

2.2 Request Handling

- Request schema handling aligned with Payment API v2 (schema `payment-request/2.0`).
- Validation error messages now list every failing field instead of only the first one.

The schema name used by 2.4.0 is payment-request/2.0, the schema defined in the Payment API Specification v2.0.

2.3 Observability

- New metric `payment_validation_failures_total`, labelled by error code.

Property	Value
Name	payment_validation_failures_total
Label	error code
Added in	payment-service 2.4.0

Table 2-2. New metric.

Example of how the metric appears, one series per error code (illustrative values):

Example metric output after a fresh start

```
payment_validation_failures_total{error_code="PAY-400"}  0
payment_validation_failures_total{error_code="PAY-401"}  0
```

3. Validation Error Messages

Before 2.4.0, a validation error message named only the first failing field. From 2.4.0, the message lists every failing field. The error code itself does not change: request validation failures are still reported with the same error code as before.

3.1 What Changes for the Reader of a Message

	Before 2.4.0	From 2.4.0
Fields named in the message	Only the first failing field	Every failing field
Error code	Unchanged	Unchanged

3.2 Example

The examples use a request that is missing two required fields, amount_minor and currency. Values are illustrative.

Before 2.4.0: only the first failing field is named

```
Request validation failed: required field 'amount_minor' is missing
(schema=payment-request/2.0)
```

From 2.4.0: every failing field is named

```
Request validation failed: required field 'amount_minor' is missing;
required field 'currency' is missing (schema=payment-request/2.0)
```

3.3 Why It Helps

When a request has more than one problem, one message now shows all of them. Developers no longer need to fix one field, resend the request, and discover the next failing field.

4. Compatibility

- No breaking changes to the public payment API are expected.
- No database migrations.

Question	Answer from these release notes
Are breaking changes to the public payment API expected?	No breaking changes are expected.
Does 2.4.0 contain database migrations?	No database migrations.
Which request schema does 2.4.0 use?	payment-request/2.0 (Payment API v2).

Table 4-1. Compatibility summary.

4.1 Impact by Component

2.4.0 changes only payment-service. The table shows what the release notes state for each part of ShopSphere.

Component	Changed in this release?	Notes
payment-service	Yes, 2.3.1 to 2.4.0	Dependency updates, error messages, new metric.
order-service (client of payment-service)	No	No breaking changes to the public payment API are expected.
api-gateway	No	-
inventory-service	No	-
postgres	No	No database migrations.
redis	No	-

Table 4-2. Components and this release.

5. Known Issues

None at release time.

Issue	Status
None at release time	-

6. Deploying 2.4.0

6.1 Operator Steps

The steps below follow the Payment Service Deployment Runbook v3.1.

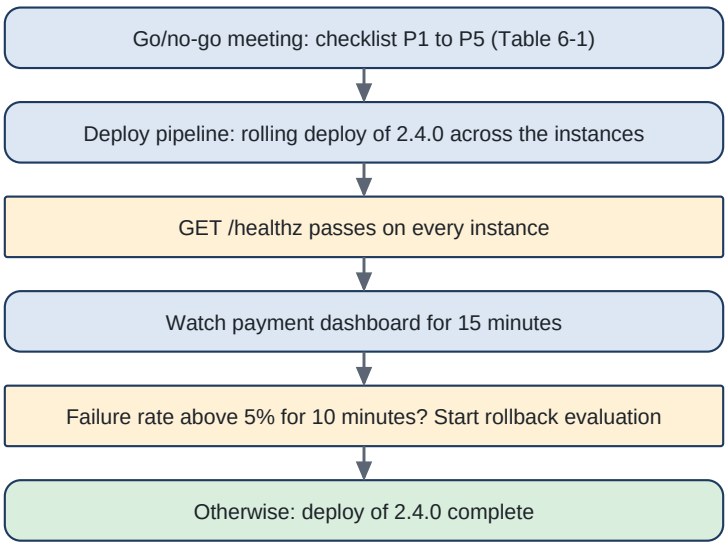

Figure 6-1. Deploying 2.4.0 with the standard runbook process.

6.2 Pre-Deployment Checklist

Deploy 2.4.0 with the normal payment-service process in the Payment Service Deployment Runbook v3.1. The runbook steps are summarised below for convenience; the runbook is the authoritative source.

Runbook step	Applied to this release
P1 Release notes reviewed in the go/no-go meeting	Review this document.
P2 Staging smoke tests passed	Check the staging smoke test run for the 2.4.0 build.
P3 Every client sends request fields the new version accepts (check dependency upgrades too)	This release upgrades schemaguard from 2.8.4 to 3.0.1 (Section 2.1).
P4 Database migrations	No database migrations (Section 4).
P5 On-call SRE aware of the deploy window	Agree the window in the go/no-go meeting.

Table 6-1. Runbook pre-deployment checklist applied to 2.4.0.

6.3 After the Rollout

Follow runbook Section 3: watch the payment dashboard for 15 minutes after the rollout completes. If the payment failure rate stays above 5% for 10 minutes, start rollback evaluation.

6.4 Rollback Target

Item	Value
Rollback target	payment-service 2.3.1 (previous version)
Database migrations to undo	None (no database migrations in 2.4.0)
Procedure	Payment Service Deployment Runbook v3.1, Section 5

6.5 Monitoring the Release

Two signals are useful after the rollout:

Signal	Type	Use
`payment_validation_failures_total`	Metric (new in 2.4.0)	Counts validation failures, labelled by error code. Look at which error codes rise after the deploy.
`PaymentFailureRateHigh`	Alert (existing)	Fires when the payment failure rate stays above 5% for 5 minutes.

Table 6-2. Signals for the release.

Example: reading the new metric (query syntax depends on your metrics tool)

```
# illustrative query - validation failures per error code, 5-minute rate
sum by (error_code) (rate(payment_validation_failures_total[5m]))
```

6.6 Testing Status

Test	Status at release time
Staging smoke tests	Passed
Known issues found	None at release time

7. Frequently Asked Questions

What kind of release is 2.4.0?

A maintenance release. It updates dependencies, improves error messages, and adds request metrics.

Which version does it replace?

2.3.1.

Which dependencies changed?

The request-validation library schemaguard (2.8.4 to 3.0.1) and the HTTP framework (latest patch version).

Is there a new metric?

Yes: payment_validation_failures_total, labelled by error code.

Do validation error messages look different?

Yes. They now list every failing field instead of only the first one.

Are breaking changes to the public payment API expected?

No breaking changes are expected.

Are there database migrations?

No.

Are there known issues?

None at release time.

8. References

Document	Version	Use
Payment API Specification	2.0	Defines schema payment-request/2.0 and the error codes.
Payment Service Deployment Runbook	3.1	How to deploy, watch and roll back payment-service.
ShopSphere Architecture Overview	1.3	Where payment-service sits in ShopSphere.

Version	Date	Owner
2.4.0	2026-10-21	Payments Team

Appendix A. Payment API v2 Request Schema

2.4.0 aligns request schema handling with Payment API v2. For convenience, this appendix repeats the field list from the Payment API Specification v2.0 (published 2026-08-14). The specification is the authoritative source.

Field	Type	Required	Description
`order_id`	`string`	yes	ShopSphere order ID, e.g. ORD-88213.
`amount_minor`	`integer`	yes	Amount in minor units (cents). 4999 = 49.99.
`currency`	`string`	yes	ISO 4217 code, e.g. EUR.
`payment_method_token`	`string`	yes	Token for the customer's payment method. Never a raw card number.
`idempotency_key`	`string`	yes	Unique per order. Reused on retries.
`customer_ref`	`string`	no	Optional customer reference.

Table A-1. Fields of payment-request/2.0 (from Payment API Specification v2.0, Section 3).

Deprecated field	Replaced by	Accepted until
`card_token`	`payment_method_token`	2027-03-31

Table A-2. Deprecated fields (from Payment API Specification v2.0, Section 4).

Appendix B. Error Codes

payment_validation_failures_total is labelled by error code. The codes of POST /v2/payments are defined in the Payment API Specification v2.0, Section 5, and repeated here.

Error code	HTTP status	Meaning
`PAY-400`	400 Bad Request	Body is not valid JSON.
`PAY-401`	422 Unprocessable Entity	Request validation failed (for example a required field is missing or has the wrong type).
`PAY-402`	402 Payment Required	Payment was declined.
`PAY-409`	409 Conflict	Idempotency key reused with a different body.
`PAY-504`	504 Gateway Timeout	Payment processor did not respond in time.

Table B-1. Error codes (from Payment API Specification v2.0).

Appendix C. Where payment-service Sits

payment-service is called by order-service and uses postgres and redis (from the ShopSphere Architecture Overview v1.3). 2.4.0 changes only payment-service; the other components are unchanged.

Figure C-1. payment-service and the components it talks to.

Appendix D. Glossary

Term	Meaning
Maintenance release	A release that updates dependencies and makes small improvements.
Breaking change	A change that makes requests that worked before stop working.
Database migration	A change to database tables that runs during a deploy.
Request-validation library	Code that checks request bodies against the request schema.
Rollback target	The version to go back to if the release has to be rolled back (2.3.1).